AI for Sustainability

The Ministry of Environment has commissioned an AI-based audit of the Delhi Airshed to identify land use patterns and pollution sources. You are provided the following datasets:

- Shapefile of the Delhi-NCR region (EPSG:4326) [File]
- Sentinel-2 RGB image patches (128×128 pixels, 10m/pixel resolution), with each image filename mapped to its center coordinates (latitude, longitude) in **.png** format. [File]
- Shapefile of the Delhi-Airshed region (EPSG:4326) [File]
- A land_cover.tif raster (ESA WorldCover 2021, 10m resolution) [File]

Your goal is to build a full pipeline involving spatial gridding, land cover classification.

Q1. Spatial Reasoning & Data Filtering [4 Marks]

- Plot the Delhi-NCR shapefile using matplotlib and overlay a 60×60 km uniform grid (2 marks)
- Filter satellite images whose center coordinates fall inside the region. (1 mark)
- Report the total number of images before and after filtering. (1 mark)

Q2. Label Construction & Dataset Preparation [6 Marks]

- For each image, extract the 128×128 corresponding land-cover patch from land_cover.tif using its center coordinate (2 marks)
- Assign the image label using the dominant (mode) land-cover class. (1 mark)
- Map ESA class codes to simplified land-use categories (e.g., Built-up, Vegetation, Water, Cropland, Others). (1 mark)
- Perform a 60/40 train-test split randomly and visualize class distribution (2 mark)

Q3. *Model Training & Supervised Evaluation* [5 Marks]

- Train a CNN model (e.g., ResNet18 or a simple custom CNN) for land-use classification. (2 marks)
- Evaluate using accuracy and F1-score. (2 marks)
- Display a confusion matrix and briefly interpret the results. (1 mark)

Supporting Information

- Image Size: 128×128 pixels
- Resolution: 10m/pixel (Sentinel-2 RGB)

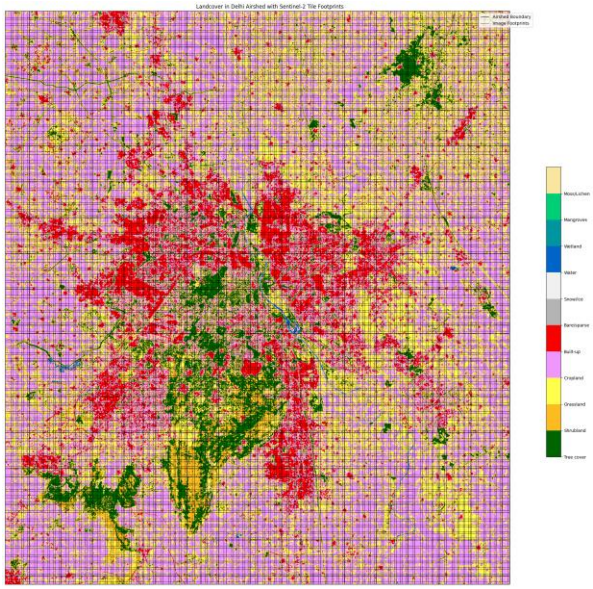
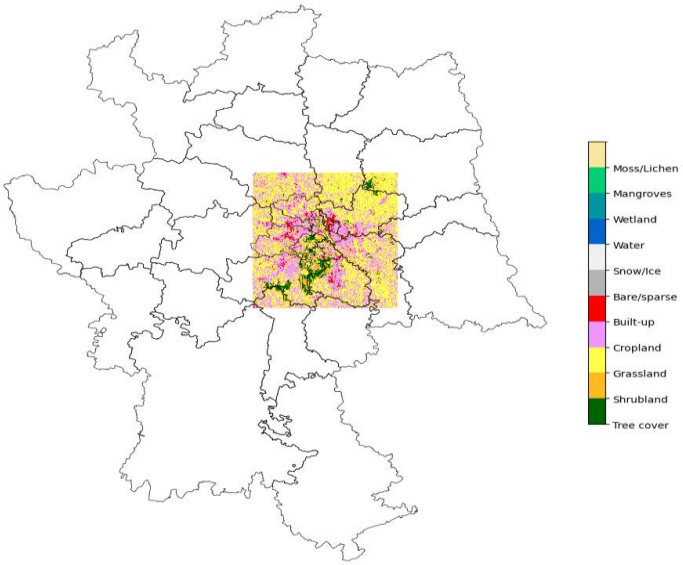
- CRS for Gridding: EPSG:32644
- Label Source: ESA WorldCover 2021 (**land_cover.tif**, 10m resolution)
- Visualization SDK: Use **geemap.Map().add_basemap("SATELLITE")**

Label Assignment Logic: Each image's label is derived from the mode value in the corresponding 128×128 label patch extracted from **land_cover.tif**, centered on its coordinate. For instance:

- 8000 pixels = class 50 (Built-up)
- 3000 pixels = class 40 (Cropland)
- 512 pixels = class 10 (Tree cover)

The assigned label would be Built-up (50), as it is the dominant class.

Helping Figures:



ESA WorldCover 2021 – Delhi NCR